MAHATMA GANDHI UNIVERSITY , KOTTAYAM
MGU-UGP (Honours)
Fifth Semester Examination
(2024 Admission Onwards)

MG5DSEMAT302 AN INVITATION TO FUZZY MATHEMATICS
Duration : 2 hours **Maximum Marks :70**

(Students should attempt at least one question from each course outcome to enhance their overall attainability)

Part A
Answer any Five questions Each question carries 2 marks

1.Define Cartesian product of two sets. [U][1]

2.Define extension principle for Fuzzy sets. [R][1]

3.Write about the axiomatic skeleton for fuzzy intersections. [U][2]

4.Write Formula for finding new t - norm if a t - norm is given. [R][2]

5.Define Reflexive , Ir reflexive and Anti - reflexive relations [U][3].

6.Define α − Compatibility class. [R][3]

7.Define a connected Graph. [U][4]

8.Let V =\{ a,b,c \}. Define Fuzzy sets σ on V and μ on E = \{ ab, bc, ac \} so that G = (V, σ , μ) is a Fuzzy Graph. [A][4]

Part B
Answer any five Questions, Each question carries 6 marks

9. Prove that a fuzzy set A on R is convex iff $A(\lambda x_1 + (1 - \lambda) x_2) \geq \min \{A(x_1), A(x_2)\}$ for all $x_1 , x_2 \in R$ and all $\lambda \in [0,1]$, where min denotes the minimum operator [A][1]

10.Is standard Fuzzy complement cutworthy ? Justify [An][1]

11.Define equilibrium point in Fuzzy complement and prove that every fuzzy complement has atmost one equilibrium [A][2]

12.Write pseudo - inverse of decreasing generator and find pseudo - inverse for $f(a) = 1 - a^p$, for any $a \in [0,1]$, $p > 0$ [A][2]

13.Show that the relation R(X,X) = \{ (x,y) : x and y have same reminder when divided by 3\} is an equivalence relation. [A][3]

14. Explain domain , range and height of a fuzzy relation R(X,Y) with equations. [U]]3]

15.Let G = (V ,σ , μ) be a Fuzzy graph . Define the following.

a) Spanning Fuzzy subgraph of G

b) Threshold graph of G corresponding to t , where $0 \leq t \leq 1$ [A][4]

16.Characterize Fuzzy cut vertices and Fuzzy bridges of Fuzzy graphs using maximum spanning trees [A][4]

Part C
Answer any Three Questions, Each question carries 10 marks

17.State and Prove second decomposition theorem in Fuzzy sets. [A][1]

18. a) Prove that equilibrium of any fuzzy complement is its own dual point.

b) Prove that for each $a \in [0,1]$, $d_a = c(a)$ if and only if $c(c(a)) = a$ [A][2]

19. Write $u_{max}(a,b)$ and prove that $\max(a,b) \leq u(a,b) \leq u_{max}(a,b)$ for all $a,b \in [0,1]$ [A][2]

20.a) Define crisp relation and characteristic function

b) Define Fuzzy relation

c) Let R be a relation among the three sets X = {English, French}

Y = { Dollar, Pound,Franc, Mark}

Z = {US,France,Britain,Germany}

Such that R = { (English, Dollar, US) , (French, Franc, France) , (English, Dollar, Canada), (French, Dollar, Canada), (English, Pound, Britain)}. Represent this Relation using 3 - dimensional membership array. [An][3]

21.Let $G = (\sigma, \mu)$ be a Fuzzy graph. Prove that xy is a Fuzzy bridge if and only if xy is not the weakest edge of any cycle. [A][4]

22. If $G = (\sigma, \mu)$ is a complete Fuzzy graph , Prove that $\mu^{\infty} = \mu$.

Is the converse true ? Justify. [An][4]

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